

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398654
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Hartnagel; Brett et al.

Turbine assembly with case-integrated cooled clearance measurement system

Abstract

A turbine assembly includes a bladed rotor mounted for rotation about an axis of the gas turbine engine, a case assembly, and a tip clearance system. The tip clearance system includes a tip clearance sensor located in an annular plenum defined between an inner case and an outer case included in the case assembly. The tip sensor is configured to monitor a tip clearance formed between the bladed rotor and the case assembly during operation of the gas turbine engine.

Inventors: Hartnagel; Brett (Indianapolis, IN), Snyder; Brandon R. (Indianapolis, IN), Pfeil; Trevor C. (Indianapolis, IN), Little; Rex M. (Indianapolis, IN)

Applicant: Rolls-Royce North American Technologies Inc. (Indianapolis, IN)

Family ID: 1000008778346

Assignee: Rolls-Royce North American Technologies Inc. (Indianapolis, IN)

Appl. No.: 18/372435

Filed: September 25, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250101876 A1	Mar. 27, 2025

Publication Classification

Int. Cl.: F01D11/24 (20060101); F01D17/02 (20060101); G01B21/16 (20060101)

U.S. Cl.:

CPC **F01D11/24** (20130101); **F01D17/02** (20130101); **G01B21/16** (20130101); F05D2220/32 (20130101); F05D2240/55 (20130101); F05D2260/20 (20130101); F05D2260/80 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7207769	12/2006	Tanioka	N/A	N/A
7414413	12/2007	Crum	N/A	N/A
7540704	12/2008	Shang	415/118	F01D 11/06
7554324	12/2008	Gualtieri	324/207.16	F01D 17/02
7922441	12/2010	Shang	N/A	N/A
8322973	12/2011	Shang	N/A	N/A
8773115	12/2013	Phillips et al.	N/A	N/A
8773155	12/2013	Yang	N/A	N/A
8827630	12/2013	Shang	N/A	N/A
8894358	12/2013	Bacic	N/A	N/A
10393149	12/2018	Mondal	N/A	N/A
11225880	12/2021	Shoemaker	N/A	N/A
2005/0088171	12/2004	Gualtieri	324/600	G01B 7/14
2011/0188994	12/2010	Shang	415/118	F01D 11/20
2014/0064924	12/2013	Warren	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111043949	12/2020	CN	N/A

OTHER PUBLICATIONS

Yu et al., A review of blade tip clearance-measuring technologies for gas turbine engines, Measurement and Control, Mar. 2020, vol. 53(3-4) 339-357. cited by applicant

Primary Examiner: Flores; Juan G

Attorney, Agent or Firm: Barnes & Thornburg

Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to gas turbine engines, and more specifically to rotor tip gap systems.

BACKGROUND

(2) Gas turbine engines are used to power aircraft, watercraft, power generators, and the like. Gas

turbine engines typically include a compressor, a combustor, and a turbine. The compressor compresses air drawn into the engine and delivers high pressure air to the combustor. In the combustor, fuel is mixed with the high pressure air and is ignited. Products of the combustion reaction in the combustor are directed into the turbine where work is extracted to drive the compressor and, sometimes, an output shaft. Left-over products of the combustion are exhausted out of the turbine.

(3) Compressors and turbines typically include alternating stages of static vane assemblies and rotating wheel assemblies. The rotating wheel assemblies include disks carrying blades around their outer edges. When the rotating wheel assemblies turn, tips of the blades which may incorporate a shroud move in close proximity to blade tracks that are attached to, or incorporated into the inner diameter of the turbine case arranged around the rotating wheel assemblies.

(4) During operation, the tips of the blades included in the rotating wheel assemblies move inwardly and outwardly relative to a centerline of the engine due to changes in centrifugal force and temperatures experienced by the blades and the wheel. Similarly, the case and blade tracks may also move inwardly and outwardly relative to a center axis of the gas turbine engine due to changes in temperature during engine operation. Because of this movement inwardly and outwardly relative to the centerline, the case around the blades may be designed to minimize clearance between the blade tips and the blade tracks. This clearance may allow combustion products to pass over the blade tips without pushing the blades, thereby contributing to lost performance within a gas turbine engine.

SUMMARY

(5) The present disclosure may comprise one or more of the following features and combinations thereof.

(6) A turbine assembly adapted for use in a gas turbine engine may include a bladed rotor, a case assembly, and a cooled tip clearance measurement system. The bladed rotor may be mounted for rotation about an axis of the gas turbine engine. The case assembly may extend circumferentially around the bladed rotor.

(7) In some embodiments, the case assembly includes an inner case and an outer case. The inner case may extend circumferentially around the bladed rotor to define an outer boundary of a gas path of the turbine assembly to block combustion products from moving through the gas path of the turbine assembly without interaction with blades on the bladed rotor. The outer case may extend circumferentially around the inner case. The outer case may be spaced radially outward of the inner case to define an annular cooling plenum therebetween. The annular cooling plenum may be configured to be flooded with cooling air at a temperature lower than a temperature of the combustion products.

(8) In some embodiments, the cooled tip clearance measurement system may include the annular cooling plenum defined between the inner case and the outer case and a tip clearance sensor located in the annular cooling plenum to engage the inner case radially outward of the gas path of the turbine assembly. The tip clearance sensor may be configured to monitor a tip clearance formed between the bladed rotor and the inner case during operation of the gas turbine engine.

(9) The tip clearance sensor may include a sensor housing and a plurality of housing stand-offs. The sensor housing may define a radially-inwardly facing surface. The plurality of housing stand-offs may extend from the radially-inwardly facing surface of the sensor housing and engage the inner case to space the radially-inwardly facing surface apart from the inner case to define a gap therebetween. The gap may allow the flow of cooling air to flow between the inner case and at least a portion of the radially-inwardly facing surface of the sensor housing.

(10) In some embodiments, the inner case may be formed to define a cooling hole that extends radially through the inner case. The cooling hole may open to the gap and the gas path so that the gap is in fluid communication with the gas path.

(11) In some embodiments, the sensor housing may include a thermal barrier coating layer. The

thermal barrier coating layer may define the radially-inwardly facing surface of the sensor housing. In some embodiments, the plurality of stand-offs may each include a thermal barrier coating layer that engages the inner case.

(12) In some embodiments, the tip clearance sensor may include a sensor housing that defines a radially-inwardly facing surface that directly engages the inner case. The sensor housing may include a thermal barrier coating layer that defines the radially-inwardly facing surface of the sensor housing.

(13) In some embodiments, the tip clearance sensor may further include a plurality of protrusions that extend away from the sensor housing into the annular cooling plenum. The plurality of protrusions may be configured to increase the surface area in contact with, and induce turbulence in the flow of cooling air supplied to the annular cooling plenum between the outer case and the inner case during operation of the gas turbine engine such that heat is more effectively transferred from the tip clearance sensor to the flow of cooling air.

(14) According to another aspect of the present disclosure, a gas turbine engine may include a compressor configured to compress air drawn in to the gas turbine engine and discharge pressurized air, a combustor configured to mix fuel with the pressurized air from the compressor and ignites the fuel to produce hot, high pressure combustion products, and a turbine assembly configured to receive the combustion products and to extract mechanical work from the combustion products as the combustion products move through the turbine assembly. The turbine assembly may include a bladed rotor mounted for rotation about an axis of the gas turbine engine, a case assembly that extends circumferentially around the bladed rotor, and a cooled tip clearance measurement system.

(15) In some embodiments, the case assembly may include an inner case and an outer case. The inner case may extend circumferentially around the bladed rotor. The outer case may extend circumferentially around the inner case. The outer case may be spaced radially outward of the inner case to define an annular cooling plenum therebetween.

(16) In some embodiments, the cooled tip clearance measurement system may include the annular cooling plenum defined between the inner case and the outer case, a tip clearance sensor located in the annular cooling plenum to engage the inner case, and a cooling air source in fluid communication with the annular cooling plenum to supply a flow of cooling air to the annular cooling plenum to transfer heat from the tip clearance sensor to the flow of cooling air so as to cool the tip clearance sensor during operation of the gas turbine engine. The tip clearance sensor may be configured to monitor a tip clearance formed between the bladed rotor and the inner case.

(17) In some embodiments, the tip clearance sensor may include a sensor housing and a plurality of housing stand-offs. The sensor housing may define a radially-inwardly facing surface. The plurality of housing stand-offs may extend from the radially-inwardly facing surface of the sensor housing and engage the inner case to space the radially-inwardly facing surface apart from inner case to define a gap therebetween. The gap may allow the flow of cooling air to flow between the inner case and the radially-inwardly facing surface of the sensor housing.

(18) In some embodiments, the inner case may be formed to define a cooling hole that extends radially through the inner case. The cooling hole may open to the gap and a gas path of the gas turbine engine so that the gap is in fluid communication with the gas path.

(19) In some embodiments, the sensor housing may include a thermal barrier coating layer. The thermal barrier coating layer may define the radially-inwardly facing surface of the sensor housing. In some embodiments, the plurality of stand-offs may each include a thermal barrier coating layer that engages the inner case.

(20) In some embodiments, the tip clearance sensor may include a sensor housing that defines a radially-inwardly facing surface that directly engages the inner case. The sensor housing may include a thermal barrier coating layer that defines the radially-inwardly facing surface of the sensor housing.

- (21) In some embodiments, the tip clearance sensor may further include a plurality of protrusions. The plurality of protrusions may extend away from the sensor housing into the annular cooling plenum.
- (22) In some embodiments, the cooling air source may be air from the compressor. In some embodiments, the cooling air source may be from another section of the engine.
- (23) According to another aspect of the present disclosure, a method may include providing a turbine assembly adapted for use in a gas turbine engine. The turbine assembly may include a bladed rotor mounted for rotation about an axis of the gas turbine engine, a case assembly, and a cooled tip clearance measurement system.
- (24) In some embodiments, the case assembly may include an inner case and an outer case. The inner case may extend circumferentially around the bladed rotor. The outer case may extend circumferentially around the inner case. The outer case may be spaced radially outward of the inner case to define an annular cooling plenum therebetween.
- (25) In some embodiments, the cooled tip clearance measurement system may include the annular cooling plenum defined between the inner case and the outer case, a tip clearance sensor located in the annular cooling plenum to engage the inner case, and a cooling air source in fluid communication with the annular cooling plenum to supply a flow of cooling air to the annular cooling plenum.
- (26) In some embodiments, the method may further include monitoring a tip clearance formed between the bladed rotor and the inner case during operation of the gas turbine engine. The method may further include conducting the flow of cooling air into the annular cooling plenum from the cooling air source to transfer heat from the tip clearance sensor to the flow of cooling air so as to cool the tip clearance sensor during operation of the gas turbine engine.
- (27) In some embodiments, the tip clearance sensor may include a sensor housing that defines a radially-inwardly facing surface and a plurality of housing stand-offs. The plurality of housing stand-offs may extend from the radially-inwardly facing surface of the sensor housing and engage the inner case to space the radially-inwardly facing surface apart from inner case to define a gap therebetween. The gap may allow the flow of cooling air to flow between the inner case and the radially-inwardly facing surface of the sensor housing.
- (28) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cutaway perspective view of a gas turbine engine that includes a fan, a compressor, a combustor, and a turbine, the turbine including a turbine section comprising a plurality of bladed rotor assemblies configured to rotate about a central axis of the gas turbine engine and a plurality of static vane assemblies in between each bladed rotor assembly;
- (2) FIG. 2 is a cross-sectional view of a portion of the turbine section of the gas turbine engine of FIG. 1 showing the turbine section further includes a case assembly having an inner case that extends circumferentially around the bladed rotor assemblies and static vanes assemblies to define an outer boundary of a gas path of the gas turbine engine and an outer case that extends circumferentially around the inner case radially outward of the inner case and a cooled tip clearance measurement system configured to monitor the blade tip clearance between the bladed rotor and the case assembly;
- (3) FIG. 3 is a cross-section view of the turbine section of FIG. 2 showing the cooled tip clearance measurement system includes an annular cooling plenum defined between the inner case and the outer case, a tip clearance sensor located in the annular cooling plenum to engage the inner case,

and a cooling air source in fluid communication with the annular cooling plenum to supply a flow of cooling air to the annular cooling plenum to transfer heat from the sensor to the cooling air so as to cool the sensor during operation of the gas turbine engine, and further showing the sensor includes a sensor housing that defines a radially-inwardly facing surface and a plurality of housing stand-offs that extend from the radially-inwardly facing surface of the sensor housing and engage the inner case to space the radially-inwardly facing surface apart from inner case to define a gap therebetween to allow the flow of cooling air to flow between the inner case and the radially-inwardly facing surface of the sensor housing and out through a cooling hole formed in the inner case into the gas path;

(4) FIG. 4 is another embodiment of a cooled tip clearance measurement system adapted for use in the turbine section of FIG. 2 showing the inner case does not include a cooling hole like in FIG. 3;

(5) FIG. 5 is another embodiment of a cooled tip clearance measurement system adapted for use in the turbine section of FIG. 2 showing the sensor housing includes a thermal barrier coating layer that defines the radially-inwardly facing surface of the sensor housing.

(6) FIG. 6 is another embodiment of a cooled tip clearance measurement system adapted for use in the turbine section of FIG. 2 showing the sensor housing includes a first thermal barrier coating layer that defines the radially-inwardly facing surface of the sensor housing and second thermal barrier coating layers on the plurality of stand-offs;

(7) FIG. 7 is another embodiment of a cooled tip clearance measurement system adapted for use in the turbine section of FIG. 2 showing the sensor includes a sensor housing that defines a radially-inwardly facing surface that directly engages the inner case and a plurality of protrusions that extend away from the sensor housing into the annular cooling plenum to induce turbulence in the flow of cooling air supplied to the annular cooling plenum between the outer case and the inner case during operation of the gas turbine engine such that heat is more effectively transferred from the sensor to the cooling air; and

(8) FIG. 8 is another embodiment of a cooled tip clearance measurement system adapted for use in the turbine section of FIG. 2 showing the sensor housing includes a thermal barrier coating layer that defines the radially-inwardly facing surface of the sensor housing that directly engages the inner case.

DETAILED DESCRIPTION OF THE DRAWINGS

(9) For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to a number of illustrative embodiments illustrated in the drawings and specific language will be used to describe the same.

(10) An illustrative aerospace gas turbine engine **10** includes a fan **12**, a compressor **14**, a combustor **16**, and a turbine **18** as shown in FIG. 1. The fan **12** is driven by the turbine **18** and provides thrust for propelling an air vehicle. The compressor **14** compresses and delivers air to the combustor **16**. The combustor **16** mixes fuel with the compressed air received from the compressor **14** and ignites the fuel. The hot, high-pressure products of the combustion reaction in the combustor **16** are directed into the turbine **18** to cause the turbine **18** to rotate about an axis **11** and drive the compressor **14** and the fan **12**. In some embodiments, the fan **12** may be replaced with a propeller, drive shaft, or other suitable configuration.

(11) The turbine **18** includes a case assembly **20**, static turbine vane assemblies **22**, rotating bladed rotor assemblies **24** between adjacent turbine vane assemblies **22**, and a cooled tip clearance measurement system **26** as shown in FIG. 2. The case assembly **20** extends circumferentially around the axis **11** of the gas turbine engine **10**. Each bladed rotor assembly **24** has a rotor **28** mounted for rotation about the axis **11** of the gas turbine engine **10** and a plurality of blades **30** coupled to the rotor **28** for rotation therewith. The cooled tip clearance measurement system **26** is configured to monitor the blade tip clearance **T** as shown in FIG. 3 between the bladed rotor **24** and the case assembly **20**.

(12) The case assembly **20** includes an inner case **32** and an outer case **34** as shown in FIGS. 2 and

3. The inner case **32** extends circumferentially around the bladed rotor **24** to define an outer boundary of a gas path **25** of the turbine assembly **18**. The inner case **32** extends circumferentially around the bladed rotor **24** to block combustion products from moving through the gas path **25** of the turbine assembly **18** without interaction with the blades **30** on the bladed rotor **24**. The outer case **34** extends circumferentially around the inner case **32** radially outward of the inner case **32**.

(13) The cooled tip clearance measurement system **26** includes an annular cooling plenum **40** and a tip clearance sensor **42** as shown in FIGS. 2 and 3. The annular cooling plenum **40** is defined between an inner case **32** and an outer case **34** included in the case assembly **20**. The tip clearance sensor **42** is located in the annular cooling plenum **40** to engage the inner case **32** radially outward of the gas path **25** of the turbine assembly **18**.

(14) The annular cooling plenum **40** is configured to be flooded with cooling air to cool the sensor **42**. The cooling air supplied to the annular cooling plenum **40** has a temperature lower than the temperature of the combustion products from the combustor **16**. The cooling air transfers heat from the sensor **42** to the cooling air so as to cool the sensor during operation of the gas turbine engine. In the illustrative embodiment, the cooled tip clearance measurement system **26** includes a cooling air source **44** in fluid communication with the annular cooling plenum **40** to supply a flow of cooling air to the annular cooling plenum **40**.

(15) The distance between the tips of the blades included in the bladed rotor **24** and the surrounding case may be important to gas turbine engine performance. Sensors **42** may be used to monitor the tip clearance **T** during engine operation. Sensors **42** may also be incorporated into an active control system which may measure and adjust the turbine tip clearance **T** for optimum performance.

(16) The sensor **42** is configured to monitor a tip clearance **T** formed between the bladed rotor **24** and the inner case **32** during operation of the gas turbine engine **10**. To provide an accurate measurement, the sensor **42** needs to remain in contact with an inner case **32** of the case assembly **20**, which carries the blade track. Temperatures in this region of the engine **10** may exceed the capability of electronic or magnetic sensor components; therefore, the sensor **42** may need to be cooled.

(17) The sensor **42** includes a sensor housing **46** and a plurality of stand-offs **48** as shown in FIG. 3. The sensor housing **46** houses the other sensor components. The sensor housing **46** defines a radially-inwardly facing surface **50**. The plurality of housing stand-offs **48** that extend from the radially-inwardly facing surface **50** of the sensor housing **46** and engage the inner case **32** to space the radially-inwardly facing surface **50** apart from inner case **32** to define a gap **52** therebetween. The gap **52** between the radially-inwardly facing surface **50** and the inner case **32** to allow the flow of cooling air to flow between the inner case **32** and the radially-inwardly facing surface **50** of the sensor housing **46**.

(18) The plurality of housing stand-offs **48** maintain contact with the inner case **32** of the case assembly **20** as shown in FIG. 3. The stand-offs **48** engage a radially-outwardly facing surface **33** of the inner case **32**.

(19) In the illustrative embodiment, the plurality of housing stand-offs **48** are castellated as shown in FIGS. 2 and 3. The plurality of housing stand-offs **48** are spaced apart from one another or castellated to define openings to the gap **52** so that the cooling air may flow between the inner case **32** and at least a portion of the radially-inwardly facing surface **50** of the sensor housing **46**.

(20) Turning back to the turbine section **18**, the cooling air source **44** is a portion of the compressed air from the compressor **14**. The cooling air is taken from the compressor **14** prior to discharge into the combustor **16** and delivered to the turbine **18** to cool the components in the turbine section **18**. The flow of cooling air may flow through cooling passageways in the turbine vanes **22** and/or the bladed rotor assembly **24** to cool the components of the turbine **18** as shown in FIG. 2.

(21) The flow of cooling air from the cooling air source **44** may flow from the annular cooling plenum **40** through cooling passageways **27** as shown in FIG. 2. Additionally, the cooling air may flow radially inward through the turbine vanes **22** and then radially outward through the bladed

rotor **24** toward the inner case **32** of the case assembly **20**.

(22) The cooling air from the cooling air source **44**, i.e. the compressed air from the compressor **14**, is cooler and at a higher pressure than the air flowing through the gas path **25** of the turbine section **18**.

(23) In the illustrative embodiment, the inner case **32** of the case assembly **20** includes a cooling hole **54** as shown in FIG. **3**. The cooling hole **54** extends radially through the inner case **32**. The cooling hole **54** opens to the gap **52** and the gas path **25** so that the gap **52** is in fluid communication with the gas path **25** as shown in FIG. **3**. In this way, the pressure differential across the inner case **32** may be used to draw the flow of cooling air into the gap **52** underneath at least a portion of the sensor housing **46**. The airflow will cool the radially-inwardly facing surface **50** of the sensor housing **46** by convection.

(24) In some embodiments, the radially-inwardly facing surface **50** of the sensor housing **46** includes heat transfer features, such as protrusions, dimples, fins, pins, or other similar features to increase the efficiency of the heat transfer. In some embodiments, like in FIG. **7**, the sensor housing **46** includes heat transfer features on other areas.

(25) Turning again to the case assembly **20**, the outer case **34** includes an outer case wall **36** and an outer case boss **38** as shown in FIG. **3**. The outer case wall **36** extends circumferentially at least partway about the axis **11**. The outer case boss **38** extends radially outward from the outer case wall **36**.

(26) In the illustrative embodiment, an opening **37** extends radially through the outer case wall **36** and the outer case boss **38** and is open to the annular cooling plenum **40**. The sensor **42** extends into the opening **37** in the illustrative embodiment.

(27) In the illustrative embodiment, the cooled tip clearance measurement system **26** includes a sensor mount **70** that couples to the outer case **34** to close the opening **37**. The sensor **42** extends through the sensor mount **70** in the illustrative embodiment.

(28) In the illustrative embodiment, the cover plate **60** is coupled to the outer case **34** to close the opening **37**. The sensor **42** extends through the opening **37** in the outer case **34**, the cover plate **60**, and the sensor mount **70**.

(29) A method of monitoring and controlling the tip clearance during engine operation may include several steps. The method includes using the tip clearance sensor **42** to monitor the tip clearance **T** formed between the blades **30** on the bladed rotor **24** and the inner case **32** during operation of the gas turbine engine **10**. While monitoring the tip clearance, the method further includes conducting the flow of cooling air into the annular cooling plenum **40** from the cooling air source **44** to transfer heat from the sensor **42** to the cooling air so as to cool the sensor **42** during operation of the gas turbine engine **10**.

(30) Another embodiment of a turbine assembly **218** in accordance with the present disclosure is shown in FIG. **4**. The turbine assembly **218** is substantially similar to the turbine assembly **18** shown in FIGS. **1-3** and described herein. Accordingly, similar reference numbers in the **200** series indicate features that are common between the turbine assembly **18** and the turbine assembly **218**. The description of the turbine assembly **18** is incorporated by reference to apply to the turbine assembly **218**, except in instances when it conflicts with the specific description and the drawings of the turbine assembly **218**.

(31) The turbine assembly **218** includes a case assembly **220**, a bladed rotor **224**, and a cooled tip clearance measurement system **226** as shown in FIG. **4**. The case assembly **220** includes an inner case **232** that extends circumferentially around the bladed rotor **224** and an outer case **234** extends circumferentially around the inner case **232** radially outward of the inner case **232**.

(32) The cooled tip clearance measurement system **226** includes an annular cooling plenum **240**, a tip clearance sensor **242**, and a cooling air source **244** as shown in FIG. **4**. The annular cooling plenum **240** is defined between the inner case **232** and the outer case **234** included in the case assembly **220**. The tip clearance sensor **242** is located in the annular cooling plenum **240** to engage

the inner case **232**. The cooling air source **244** is in fluid communication with the annular cooling plenum **240** to supply a flow of cooling air to the annular cooling plenum **240** to transfer heat from the sensor **242** to the cooling air so as to cool the sensor during operation of the gas turbine engine. (33) The sensor **242** includes a sensor housing **246** and a plurality of stand-offs **248** as shown in FIG. **4**. The sensor housing **246** defines a radially-inwardly facing surface **250**. The plurality of housing stand-offs **248** that extend from the radially-inwardly facing surface **250** of the sensor housing **246** and engage the inner case **232** to space the radially-inwardly facing surface **250** apart from inner case **232** to define a gap **252** therebetween.

(34) The gap **252** between the radially-inwardly facing surface **250** and the inner case **232** to allow the flow of cooling air to flow between the inner case **232** and the radially-inwardly facing surface **250** of the sensor housing **246**. Unlike the inner case **32** in FIGS. **2** and **3**, the inner case **232** of the case assembly **220** does not include a cooling hole so that the gap **252** is not in fluid communication with the gap path **225**.

(35) Another embodiment of a turbine assembly **318** in accordance with the present disclosure is shown in FIG. **5**. The turbine assembly **318** is substantially similar to the turbine assembly **18** shown in FIGS. **1-3** and described herein. Accordingly, similar reference numbers in the **300** series indicate features that are common between the turbine assembly **18** and the turbine assembly **318**. The description of the turbine assembly **18** is incorporated by reference to apply to the turbine assembly **318**, except in instances when it conflicts with the specific description and the drawings of the turbine assembly **318**.

(36) The turbine assembly **318** includes a case assembly **320**, a bladed rotor **324**, and a cooled tip clearance measurement system **326** as shown in FIG. **5**. The case assembly **320** includes an inner case **332** that extends circumferentially around the bladed rotor **324** and an outer case **334** extends circumferentially around the inner case **332** radially outward of the inner case **332**.

(37) The cooled tip clearance measurement system **326** includes an annular cooling plenum **340**, a tip clearance sensor **342**, and a cooling air source **344** as shown in FIG. **5**. The annular cooling plenum **340** is defined between the inner case **332** and the outer case **334** included in the case assembly **320**. The tip clearance sensor **342** is located in the annular cooling plenum **340** to engage the inner case **332**. The cooling air source **344** is in fluid communication with the annular cooling plenum **340** to supply a flow of cooling air to the annular cooling plenum **340** to transfer heat from the sensor **42** to the cooling air so as to cool the sensor during operation of the gas turbine engine.

(38) The sensor **342** includes a sensor housing **346** and a plurality of stand-offs **348** as shown in FIG. **5**. The sensor housing **346** defines a radially-inwardly facing surface **350**. The plurality of housing stand-offs **348** that extend from the radially-inwardly facing surface **350** of the sensor housing **346** and engage the inner case **332** to space the radially-inwardly facing surface **350** apart from inner case **332** to define a gap **352** therebetween. The gap **352** between the radially-inwardly facing surface **350** and the inner case **332** to allow the flow of cooling air to flow between the inner case **332** and the radially-inwardly facing surface **350** of the sensor housing **346**.

(39) In the illustrative embodiment, the sensor housing **346** includes a thermal barrier coating layer **356** that defines the radially-inwardly facing surface **350** of the sensor housing **346** as shown in FIG. **5**. The thermal barrier coating layer **356** is configured to insulate the sensor **342** to reduce the heat conduction between inner case **332** and the sensor housing **346**.

(40) Another embodiment of a turbine assembly **418** in accordance with the present disclosure is shown in FIG. **6**. The turbine assembly **418** is substantially similar to the turbine assembly **18** shown in FIGS. **1-3** and described herein. Accordingly, similar reference numbers in the **400** series indicate features that are common between the turbine assembly **18** and the turbine assembly **418**. The description of the turbine assembly **18** is incorporated by reference to apply to the turbine assembly **418**, except in instances when it conflicts with the specific description and the drawings of the turbine assembly **418**.

(41) The turbine assembly **418** includes a case assembly **420**, a bladed rotor **424**, and a cooled tip

clearance measurement system **426** as shown in FIG. **6**. The case assembly **420** includes an inner case **432** that extends circumferentially around the bladed rotor **424** and an outer case **434** extends circumferentially around the inner case **432** radially outward of the inner case **432**.

(42) The cooled tip clearance measurement system **426** includes an annular cooling plenum **440**, a tip clearance sensor **442**, and a cooling air source **444** as shown in FIG. **6**. The annular cooling plenum **440** is defined between the inner case **432** and the outer case **434** included in the case assembly **420**. The tip clearance sensor **442** is located in the annular cooling plenum **440** to engage the inner case **432**. The cooling air source **444** is in fluid communication with the annular cooling plenum **440** to supply a flow of cooling air to the annular cooling plenum **440** to transfer heat from the sensor **442** to the cooling air so as to cool the sensor during operation of the gas turbine engine.

(43) The sensor **442** includes a sensor housing **446** and a plurality of stand-offs **448** as shown in FIG. **6**. The sensor housing **446** defines a radially-inwardly facing surface **450**. The plurality of housing stand-offs **448** that extend from the radially-inwardly facing surface **450** of the sensor housing **446** and engage the inner case **432** to space the radially-inwardly facing surface **450** apart from inner case **432** to define a gap **452** therebetween. The gap **452** between the radially-inwardly facing surface **450** and the inner case **432** to allow the flow of cooling air to flow between the inner case **432** and the radially-inwardly facing surface **450** of the sensor housing **446**.

(44) In the illustrative embodiment, the sensor housing **446** includes a first thermal barrier coating layer **456** that defines the radially-inwardly facing surface **450** of the sensor housing **446** as shown in FIG. **6**. The thermal barrier coating layer **456** is configured to insulate the sensor **442** to reduce the heat conduction between inner case **432** and the sensor housing **446**.

(45) In the illustrative embodiment, the plurality of stand-offs **448** each includes a second thermal barrier coating layer **458** as shown in FIG. **6**. The second thermal barrier coating layers **458** each engage the inner case **432**. Like the first thermal barrier coating layer **456**, the second thermal barrier coating layers **458** are configured to insulate the sensor **442** to reduce the heat conduction between inner case **432** and the sensor **442**.

(46) Another embodiment of a turbine assembly **518** in accordance with the present disclosure is shown in FIG. **7**. The turbine assembly **518** is substantially similar to the turbine assembly **18** shown in FIGS. **1-3** and described herein. Accordingly, similar reference numbers in the **440** series indicate features that are common between the turbine assembly **18** and the turbine assembly **518**. The description of the turbine assembly **18** is incorporated by reference to apply to the turbine assembly **518**, except in instances when it conflicts with the specific description and the drawings of the turbine assembly **518**.

(47) The turbine assembly **518** includes a case assembly **520**, a bladed rotor **524**, and a cooled tip clearance measurement system **526** as shown in FIG. **7**. The case assembly **520** includes an inner case **532** that extends circumferentially around the bladed rotor **524** and an outer case **534** extends circumferentially around the inner case **532** radially outward of the inner case **532**.

(48) The cooled tip clearance measurement system **526** includes an annular cooling plenum **540**, a tip clearance sensor **542**, and a cooling air source **544** as shown in FIG. **7**. The annular cooling plenum **540** is defined between the inner case **532** and the outer case **534** included in the case assembly **520**. The tip clearance sensor **542** is located in the annular cooling plenum **540** to engage the inner case **532**. The cooling air source **544** is in fluid communication with the annular cooling plenum **540** to supply a flow of cooling air to the annular cooling plenum **540** to transfer heat from the sensor **542** to the cooling air so as to cool the sensor during operation of the gas turbine engine.

(49) The sensor **542** includes a sensor housing **546** and a plurality of protrusions **560** as shown in FIG. **7**. The sensor housing **546** defines a radially-inwardly facing surface **550**. The protrusions **560** extend away from the sensor housing **546** into the annular cooling plenum **540**. The protrusions **560** are configured to increase the surface area in contact with, and induce turbulence in the flow of cooling air supplied to the annular cooling plenum **540** between the outer case **534** and the inner case **532** during operation of the gas turbine engine **10** such that heat is more effectively transferred

from the sensor **542** to the cooling air.

(50) In the illustrative embodiment, the radially-inwardly facing surface **550** of the sensor housing **546** is directly engaged with the inner case **532** as shown in FIG. 7. In some embodiments, the sensor **542** may include a plurality of stand-offs like the other embodiments that space the radially-inwardly facing surface **550** radially apart from the inner case **532**. In some embodiments, the sensor housing **546** may include a thermal barrier coating layer that defines the radially-inwardly facing surface **550**.

(51) Another embodiment of a turbine assembly **618** in accordance with the present disclosure is shown in FIG. 8. The turbine assembly **618** is substantially similar to the turbine assembly **18** shown in FIGS. 1-3 and described herein. Accordingly, similar reference numbers in the **600** series indicate features that are common between the turbine assembly **18** and the turbine assembly **618**. The description of the turbine assembly **18** is incorporated by reference to apply to the turbine assembly **618**, except in instances when it conflicts with the specific description and the drawings of the turbine assembly **618**.

(52) The turbine assembly **618** includes a case assembly **620**, a bladed rotor **624**, and a cooled tip clearance measurement system **626** as shown in FIG. 8. The case assembly **620** includes an inner case **632** that extends circumferentially around the bladed rotor **624** and an outer case **634** extends circumferentially around the inner case **632** radially outward of the inner case **632**.

(53) The cooled tip clearance measurement system **626** includes an annular cooling plenum **640**, a tip clearance sensor **642**, and a cooling air source **644** as shown in FIG. 8. The annular cooling plenum **640** is defined between the inner case **632** and the outer case **634** included in the case assembly **620**. The tip clearance sensor **642** is located in the annular cooling plenum **640** to engage the inner case **632**. The cooling air source **644** is in fluid communication with the annular cooling plenum **640** to supply a flow of cooling air to the annular cooling plenum **640** to transfer heat from the sensor **642** to the cooling air so as to cool the sensor during operation of the gas turbine engine.

(54) The sensor **642** includes a sensor housing **646** that defines a radially-inwardly facing surface **650** that directly engages the inner case **532** as shown in FIG. 8. In the illustrative embodiment, the sensor housing **646** includes a thermal barrier coating layer **662** that defines the radially-inwardly facing surface **650** of the sensor housing **646**.

(55) The thermal barrier coating layer **662** is configured to insulate the sensor **642** to reduce the heat conduction between inner case **632** and the sensor housing **646**. In the illustrative embodiment, the thermal barrier coating layer **662** directly engages the inner case **632**.

(56) The present invention is a method to cool a tip clearance sensor **42**, **242**, **342**, **442**, **542**, **642** in a gas turbine engine **10**. This invention may be used in either the compressor **14** or turbine **18** of the engine **10**; however, the example embodiment discussed here is mounted in the turbine **18**.

(57) The blades **30** are typically arranged around the outer diameter of a rotor **28**. Stages of one or more bladed discs may be coaxially assembled to form a rotor which rotates about the axis **11**. The bladed rotor **24** is housed with a static structure which may include an inner case **32** and an outer case **34**. The inner surface of the static structure which is immediately adjacent to the tips of the blades **30** and surrounds the bladed rotor **24** may be referred to as the blade track. The blade track may be part of a separate component that attaches to the inner case **32**, or it may be integral to the inner case **32** itself.

(58) The radial distance between the tips of the blades **30** and the surrounding blade track **32** is the tip clearance **T**, which may be important to gas turbine engine performance. Sensors **42** may be used to monitor the tip clearance **T** during engine operation and may also be incorporated into an active control system which can measure and adjust the turbine tip clearance **T** for optimum performance.

(59) To provide an accurate measurement, the sensor **42**, **242**, **342**, **442**, **542**, **642** needs to be in contact with the inner case **32**, **232**, **332**, **432**, **532**, **632**, which carries the blade track. Temperatures in this region of the engine **10** may exceed the capability of electronic or magnetic sensor

components.

(60) Therefore, the sensor **42, 242, 342, 442, 542, 642** may need to be cooled.

(61) Feet or stand-offs **48, 248, 348, 448** may be used on the bottom of the sensor housing **46, 246, 346, 446** to create an insulating air gap **52, 252, 352, 452** between the housing **46, 246, 346, 446** and the hot inner case **32, 232, 332, 432**.

(62) Cooling air supplied to the annular cooling plenum **40, 240, 340, 440** cools the sensor **42, 242, 342, 442** through convective heat transfer. However, the cooling benefit may be limited without motivating the airflow between the housing **46, 246, 346, 446** and the inner case **32, 232, 332, 432**. As shown in FIG. 3, the inner case **32** includes a cooling hole **54** that opens to the gap **52** and the gas path **25**. The cooling hole **54** causes the high pressure cooling air to be drawn into the gap **52** underneath the sensor housing **46**. The airflow into the gap **52** will cool the bottom of the sensor housing **46** by convection. Dimples, fins, pins, or other similar features may be employed on the radially-inwardly facing surface **50** of the sensor housing **46** to increase the efficiency of the heat transfer.

(63) In some embodiments, like in FIG. 7, the exterior of the sensor housing **546** has cooling features to facilitate the extraction of heat from inside the housing **546**. The features may be protrusions, pins, fins, dimples, turbulators, or other features which increase the efficiency of the heat transfer. The exterior of the housing **546** may be cooled by the compressor bleed or discharge air moving through the plenum **540** between the inner case **532** and the outer case **534**. The cooling air may be from another cooling air source **44, 244, 344, 444, 544, 644** such as nacelle air or external air from an air scoop or motivated by a blower.

(64) A method to isolate the sensor housing **46, 246, 346, 446** from the hot inner case **32, 232, 332, 432** uses feet or stand-offs **48, 248, 348, 448** on the bottom of the sensor housing **46, 246, 346, 446** to create an air gap **52, 252, 352, 452** between the housing **46, 246, 346, 446** and the hot inner case **32, 232, 332, 432** as shown in FIGS. 3-6. This may also reduce the heat conduction between the inner case **32, 232, 332, 432** and the sensor housing **46, 246, 346, 446** and allow the cooler cavity air to circulate through the air gap **52, 252, 352, 452**.

(65) Another method to isolate the sensor housing **346, 446, 646** from the hot inner case **332, 432, 632** uses a thermal barrier coating layer **356, 456, 458, 662** on the bottom of the sensor housing **346, 446, 646** as shown in FIGS. 5, 6, and 8. This reduces the heat conduction between the inner case **332, 432, 632** and the sensor housing **346, 446, 646**.

(66) The thermal barrier coating layer **356, 456, 458** may be employed on sensors **342, 442** with stand-offs **348, 448** that create the air gap **352, 452** between the sensor housing **346, 446** and the hot inner case **332, 432**. Further, the thermal barrier coating layer **458** may be applied to the stand-offs **448**, or as the stand-offs **448** themselves.

(67) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A turbine assembly adapted for use in a gas turbine engine, the turbine assembly comprising a bladed rotor mounted for rotation about an axis of the gas turbine engine, a case assembly including an inner case that extends circumferentially around the bladed rotor to define an outer boundary of a gas path of the turbine assembly to block combustion products from moving through the gas path of the turbine assembly without interaction with blades on the bladed rotor and an outer case that extends circumferentially around the inner case and is spaced radially outward of the inner case to define an annular cooling plenum therebetween and configured to be flooded with cooling air at a temperature lower than a temperature of the combustion products, and a cooled tip

clearance measurement system including the annular cooling plenum defined between the inner case and the outer case and a tip clearance sensor located in the annular cooling plenum to engage the inner case radially outward of the gas path of the turbine assembly and configured to monitor a tip clearance formed between the bladed rotor and the inner case during operation of the gas turbine engine, wherein the tip clearance sensor includes a sensor housing that defines a radially-inwardly facing surface and a plurality of housing stand-offs that extend from the radially-inwardly facing surface of the sensor housing and engage the inner case to space the radially-inwardly facing surface apart from the inner case to define a gap therebetween to allow a flow of the cooling air to flow between the inner case and at least a portion of the radially-inwardly facing surface of the sensor housing.

2. The turbine assembly of claim 1, wherein the inner case is formed to define a cooling hole that extends radially through the inner case and opens to the gap and the gas path so that the gap is in fluid communication with the gas path.

3. The turbine assembly of claim 2, wherein the sensor housing includes a thermal barrier coating layer that defines the radially-inwardly facing surface of the sensor housing.

4. The turbine assembly of claim 1, wherein the sensor housing includes a thermal barrier coating layer that defines the radially-inwardly facing surface of the sensor housing.

5. The turbine assembly of claim 4, wherein the plurality of stand-offs each includes a thermal barrier coating layer that engages the inner case.

6. A gas turbine engine comprising a compressor configured to compress air drawn in to the gas turbine engine and discharge pressurized air, a combustor configured to mix fuel with the pressurized air from the compressor and ignites the fuel to produce combustion products, a turbine assembly configured to receive the combustion products and to extract mechanical work from the combustion products as the combustion products move through the turbine assembly, the turbine assembly including a bladed rotor mounted for rotation about an axis of the gas turbine engine, a case assembly including an inner case that extends circumferentially around the bladed rotor to define an outer boundary of a gas path of the turbine assembly to block the combustion products from moving through the gas path of the turbine assembly without interaction with blades on the bladed rotor and an outer case that extends circumferentially around the inner case and is spaced radially outward of the inner case to define an annular cooling plenum therebetween and configured to be flooded with cooling air at a temperature lower than a temperature of the combustion products, and a cooled tip clearance measurement system including the annular cooling plenum defined between the inner case and the outer case, a tip clearance sensor located in the annular cooling plenum to engage the inner case and configured to monitor a tip clearance formed between the bladed rotor and the inner case, and a cooling air source in fluid communication with the annular cooling plenum to supply a flow of the cooling air to the annular cooling plenum to transfer heat from the tip clearance sensor to the flow of the cooling air so as to cool the tip clearance sensor during operation of the gas turbine engine, wherein the tip clearance sensor includes a sensor housing that defines a radially-inwardly facing surface and a plurality of housing stand-offs that extend from the radially-inwardly facing surface of the sensor housing and engage the inner case to space the radially-inwardly facing surface apart from the inner case to define a gap therebetween to allow the flow of the cooling air to flow between the inner case and at least a portion of the radially-inwardly facing surface of the sensor housing.

7. The gas turbine engine of claim 6, wherein the inner case is formed to define a cooling hole that extends radially through the inner case and opens to the gap and the gas path of the gas turbine engine so that the gap is in fluid communication with the gas path.

8. The gas turbine engine of claim 6, wherein the sensor housing includes a thermal barrier coating layer that defines the radially-inwardly facing surface of the sensor housing.

9. The gas turbine engine of claim 8, wherein the plurality of stand-offs each includes a thermal

barrier coating layer that engages the inner case.

10. The gas turbine engine of claim 6, wherein the cooling air source is air from the compressor.
